Health Predictor Model

Meeting note 10-27-2026

These are subjected to Change till Friday

Project Overview

The Health Predictor Model project is a student-led exercise in building a cardiovascular disease prediction system. Members will work through the machine learning process from dataset selection to model evaluation and deployment. The finished system is intended as a learning project, not a clinical diagnostic tool.

Scope and Project Goals

The project covers:

- dataset selection,
- data cleaning
- exploratory data analysis (EDA)
- Feature engineering
- model training
- model interpretation
- API
- Documentation
- Final presentation.

This will not be (Outside of Project Scope):

- Clinical diagnosis
- Medical advice or treatment recommendations
- Handling protected health information
- large-scale cloud infrastructure

Software and Tools

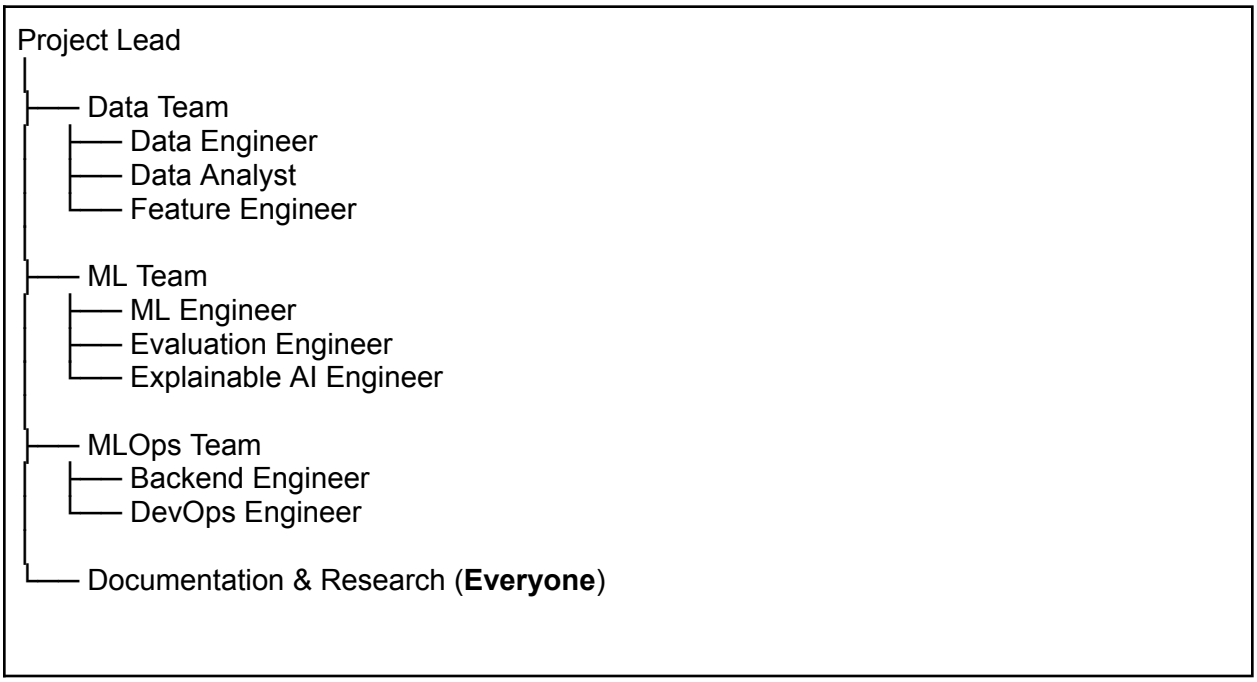
- Python; Pandas and NumPy for data preparation and numerical work.
- Matplotlib and Seaborn for charts and exploratory analysis.
- Scikit-learn for baseline models and evaluation; XGBoost and LightGBM for more advanced models.
- FastAPI for the prediction API; Docker for a reproducible deployment.
- GitHub for source control, issues, pull requests, and task tracking.

If any of you guys can think of better ways/tech stacks, convince me through discord.
Trying to use AI is also ok with me (Make sure you understand what AI spits at you).

Team Roles and Expected Work

Members will discuss their interests and experience before roles are assigned. Roles describe the main area of responsibility; members may help across workstreams as needed.

Roles (Subjected to Change by Friday)



Machine Learning Workflow

1. Data acquisition
2. Cleaning and validation
3. Exploratory data analysis
4. Feature engineering
5. Baseline models
6. Advanced models
7. Evaluation
8. Experiment tracking
9. Deployment

Project Schedule

Dates (2026) Planned work	
Oct 27 – Nov 2	Onboarding and dataset selection
Nov 3 – Nov 16	Data cleaning, validation, and exploratory analysis
Nov 17 – Nov 23	Feature engineering
Nov 24 – Nov 30	Baseline model training
Dec 1 – Dec 7	Advanced models and tuning
Dec 8 – Dec 14	Model interpretation and final evaluation
Dec 15 – Dec 20	Deployment and documentation
Dec 21 – Dec 23	Final presentation and project handoff

First Meeting discussion

- Introductions and skill inventory
- Workload and communication expectations
- Project aims and success criteria
- Machine learning workflow
- Dataset options and selection criteria

Team Communication

- Use Discord as the main team communication channel.
- Share meeting notes and agreed action items after each meeting.

GitHub rules

- Work on a feature branch; do not commit directly to main.
- Open a pull request before merging any change.
- Have at least one other member review each pull request.

Weekly updates

At each weekly check-in be ready to summarize:

- what you finished
- what you plan to do next
- any blockers
- any help you need

Final Deliverables (What we want by the end)

- Cleaned and validated dataset
- EDA report and visualizations
- Feature engineering pipeline
- Baseline and advanced model comparison report
- Selected trained model
- Model Experiment tracking records
- FastAPI prediction service and Docker configuration
- Project documentation and final presentation